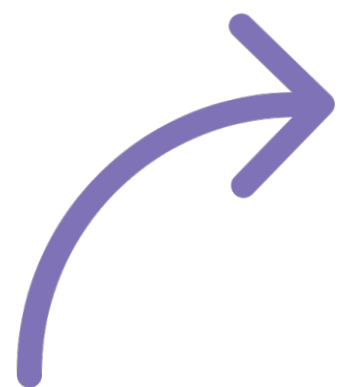


Systems Awareness

A Guide for Systems Innovators





Where am I?

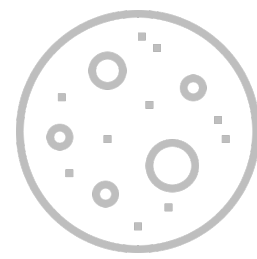
This paper forms part of our set of 20 guides covering all the systems innovation content. You are in the 2nd guide in the series in the first section on systems thinking



Systems Thinking



Systems Awareness



Systems Theory



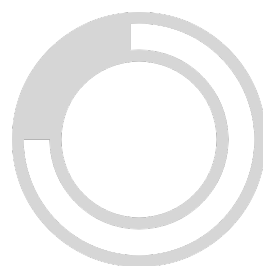
Complexity Theory



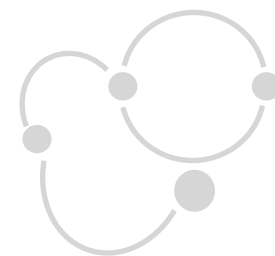
Adaptive Systems



System Inquiry



Systems Modeling



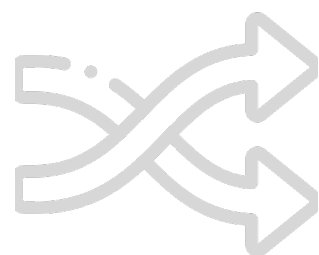
Mapping



Actor Mapping



Leverage Points



Systems Change



Horizons



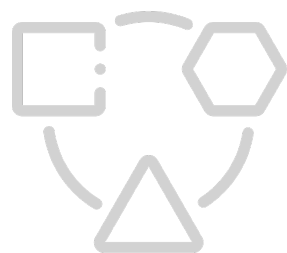
Two Loops



Multi-Level



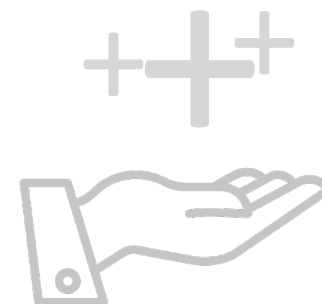
Narratives



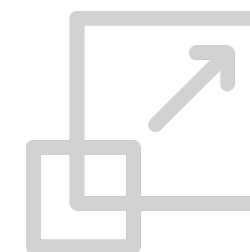
Systems Building



Networks



Value Network

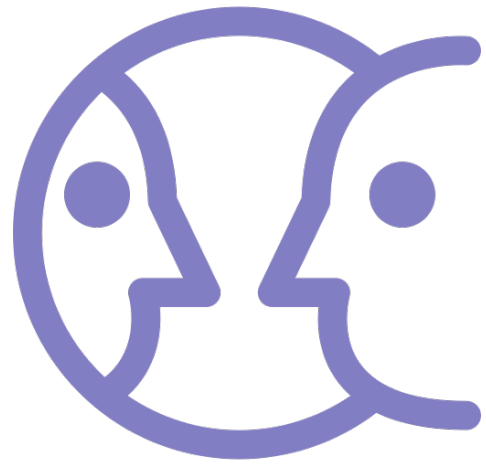


Scaling Change



Systemic Evaluation

What we Cover in this Guide



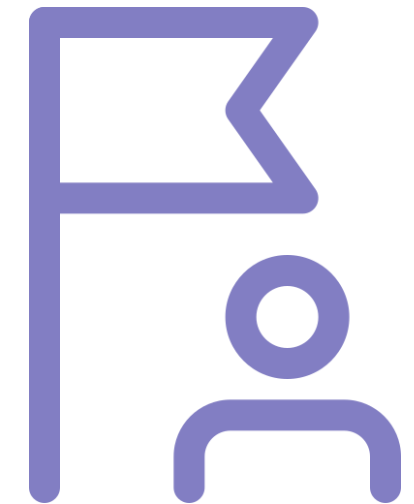
Reflexive
Thinking



Ecosystems
Thinking



Personal
Change



Systems
Leadership

Shifting Our Thinking

If we wish to achieve some kind of systems-level innovation this is going to come from a different kind of thinking, but more than this a different kind of awareness. There is nothing more true than "the same old thinking, the same old results." It follows that the same old reductionist analytical thinking focused on parts will surely result in the same old kind of changes to parts without systemic changes.

This awareness of whole systems that is the foundation for systems innovation we can call "systems awareness." It is here we will start our journey as systems innovators. The journey to systems-level innovation starts with first assessing and trying to shift our awareness and thinking from "ego-systems awareness" to "ecosystems awareness".



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to the explainer video

Awareness



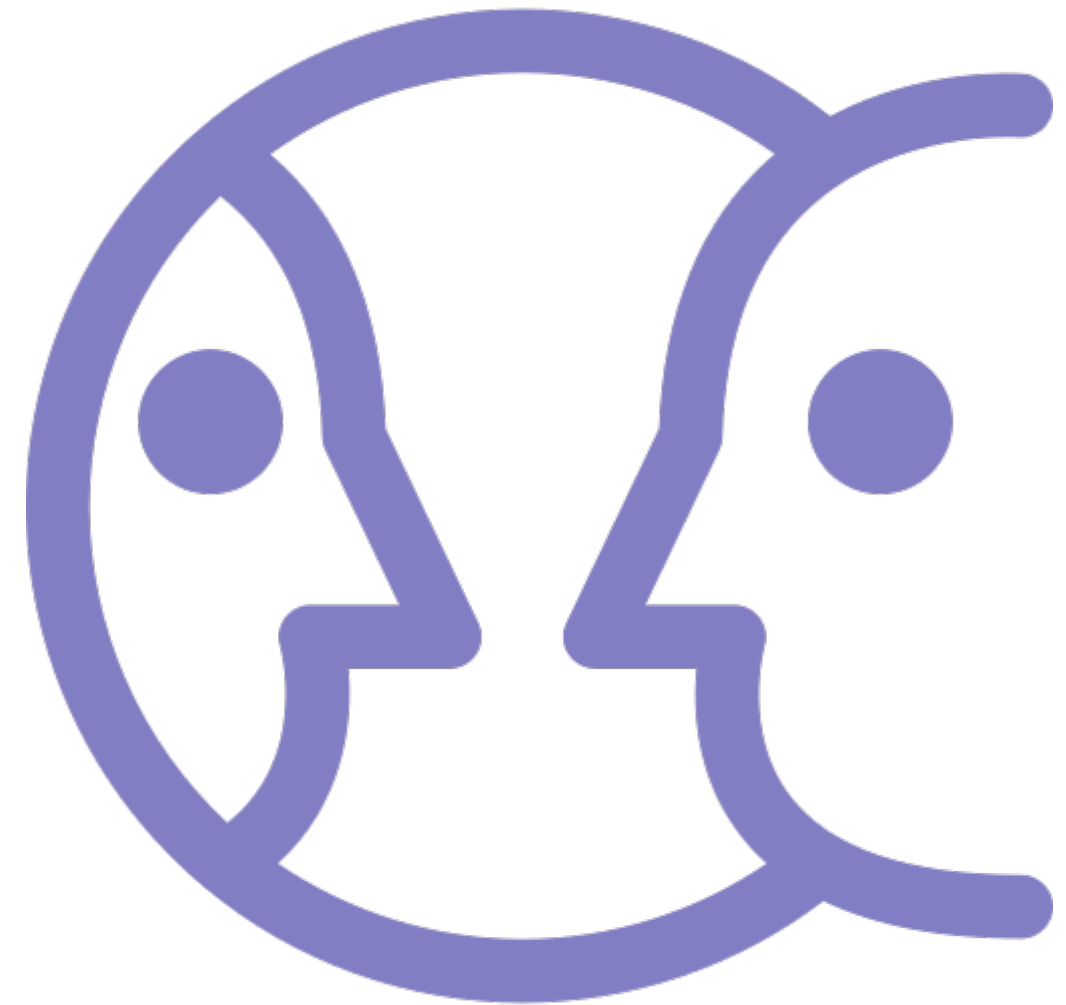
Awareness is the ability to know directly and to perceive, to feel, or to be conscious of events, objects, thoughts, emotions, or sensory patterns.^[1] Changes in our level of awareness do not happen by accident but takes a concerted effort. It requires first being self-aware of one's own ways of thinking and perspective. Awareness of one's own way of seeing the world and the process through which we reason is seen as a prerequisite to becoming an effective systems thinker.

In this sense we can understand systems thinking as a form of meta-language in that one of its explicit aims is to help individuals to understand their ways of thinking better, their underlying assumptions, processes of reasoning; how our actions and the world they create lead directly from how we reason and see the world. We call this reflexive thinking and it is from this that we can begin to start to work on shifting our thinking to a different level, to the systems level. This is what we will be covering in this guide.

"You cannot change a system unless
you transform consciousness"

- *Otto Scharmer*

Reflexive Thinking



Seeing Systems

Systems thinking is about trying to improve the quality of our thinking so as to not just see parts, but instead to see systems; the systems we form part of in our world. For most adults seeing systems does not come naturally, in our everyday way of being we see parts and draw a few connections that are very much centered around us and our particular place in the world. To see systems is to in some way overcome this self-centered view of the world.

This requires us to overcome some of the limitations, flaws, and biases in our thinking resulting from our egoism so as to see the systems we form part of independent from ourselves. Calling oneself a systems thinker should be a commitment to an ongoing learning process of examining and trying to improve our thinking to become better at seeing systems. This starts with first understanding how we see the world, and the existing limitations in our reasoning.



Why Reflexive Thinking?



"We have met the enemy and he is us."^[2] This saying from the cartoon character Pogo captures a key aspect to systems thinking, it starts with a recognition that our thinking and world-view determine how we act which leads to the world we create around us. If we want paradigm-shifting breakthroughs it will happen by changing our way of thinking and this will only happen by investing the time and energy in understanding our ways of thinking and current limitations.

If there is something systemically dysfunctional about a system we create then the real issue is likely not "out there" but instead a product of our ways of thinking. Thus starting from the beginning with systems change means an assessment of our world view, how we look at the world, this is called reflexivity. Systems thinking starts by being reflexive about our worldview and for this assessment of our ways of thinking we can use critical thinking.^[3]

What is Reflexive thinking

Reflexivity refers to circular relationships between cause and effect, in the context of thinking it refers to a bidirectional relationship between our subjective understanding of the world and the objective world we perceive. That everything we see is a function of our perception which is a product of our conceptual categories and paradigm. Our mental models shape what we see but then there is a feedback process where what we see also shapes our categories and perception.

Being reflexive means examining and being aware of the assumptions we bring into whatever we engage ourselves in. It is a recognition that none of us are, full separate and objective observers but we all hold opinions and preconceptions that are a product of what we have learned and our past experiences.^[4]



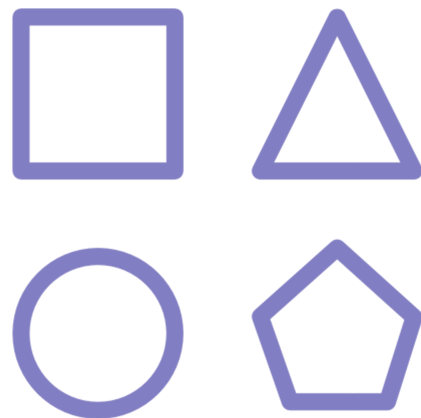


What We See

Our conceptual models
shape everything that we
see

Conceptual Models

Our paradigm shapes our
conceptual models



What We Do

How we see the world
determines how we act in it

The World We Create

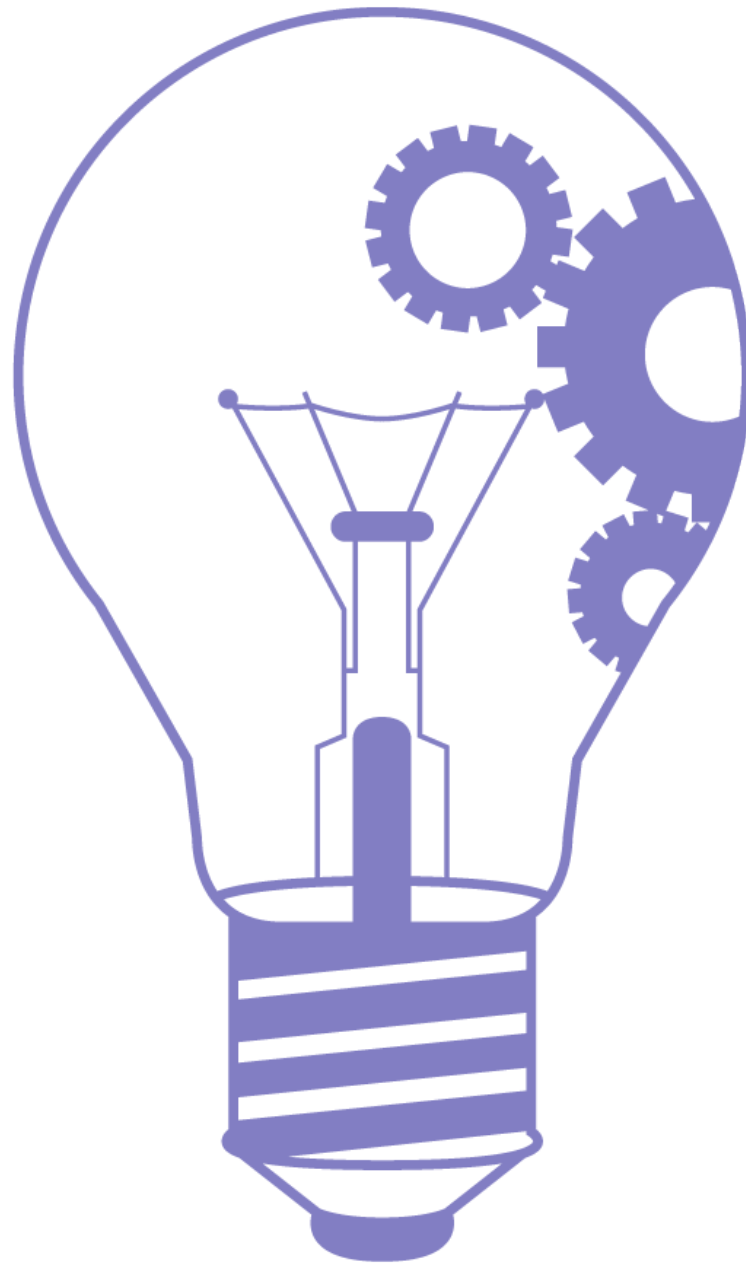
Our actions create the world we
live in which feeds back to shape
our thinking again



"You all know the saying 'seeing is believing' I would argue that other times things need to be believed in to be seen. Our narrative changes our behavior which changes reality"

- *Andrew Lo*

What is Critical Thinking?



Systems thinking before anything is about thinking, it is a type of thinking, thus we can never be good systems thinkers without first being good thinkers.

Constructive thinking does not come naturally to us, it requires a concerted effort to become aware of how we see the world, how we reason and how we might improve our reasoning.^[5]

We call this effort to improve the quality of our thinking, critical thinking. Critical thinking is about being reflective about our thinking, questioning our worldview, taking it apart to better understand it and to ensure a baseline of quality.

Critical thinking is a form of metacognition, it is self-directed and self-monitored; it is about developing the conceptual tools to be able to think for oneself.^[5]



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to the explainer video

Thinking Critically

For every human being thinking comes naturally, however typically we do not think systematically and constructively.^[6] Much of our reasoning left to itself has limitations and failings. Our thinking is often biased, unfocused, distorted, partial, uninformed, unconscious and we largely accept what we have been socialized into with only limited standards of validation placed upon it.

Like achieving high standards in any other area - such as being a great athlete or physicist - the effective use of reasoning has to be learned, cultivated and practiced and this is done through the application of standards to our thinking.^[6]

Metacognition is the activity of examining the processes by which we think about and arrive at our own beliefs. It involves analysis of one's cognitive process and adjusting them according to meta-standards. The aim is to try and improve the clarity, accuracy, relevance, depth, breadth and logical consistency of our thinking.





Thinking Quality

It is the function of the mind to create thoughts, however the quality of our thinking varies enormously from person to person.^[7] Achievement of quality requires standards of quality. Systematic thinking involves doing the work necessary to research and understand things for oneself, instead of giving the responsibility to another.

We are thinking systematically or critically when we rely on reason; require evidence; ignore no known evidence; follow evidence wherever it leads; are concerned more with finding the best explanation than being right; analyze apparent contradictions in our reasoning; ask questions and seek knowledge that is valid within many frames of reference, not just one.^[7]

Overview



Universal intellectual standards are standards that should be applied to thinking whenever one is interested in checking the quality of reasoning about a problem, issue, or situation.^[8]

To think systematically entails having command of these standards. While there are many standards The Critical Thinking Institute defines a list of the central elements of effective reasoning including: Clarity, Accuracy, Precision, Relevance, Depth, Breadth, Logic and Fairness.^[8]



Clarity

Clarity is the quality of being clear, coherent and intelligible. Clarity is a universal standard of reasoning. If a proposition is unclear, we cannot determine whether it is accurate or relevant, in fact, we cannot tell anything about it because we do not yet know what it is saying. Thinking and statements have to be refined in order to simplify and clarify them. Once an idea has been clarified the whole communications process can be greatly facilitated and more impactful.





Accuracy

Accuracy is the degree to which the result of a statement or piece of information conforms to the correct state. Try asking your self these questions to achieve better accuracy: is the statement accurate enough to deal with the topic at hand? Can we check that it is accurate? And how accurate is it, can we measure its degree of accuracy?

Precision

Precision may seem similar to accuracy however it is possible to make many statements that are clear and accurate but not precise and thus have little value. For example, saying there are many galaxies in the universe does not really inform us of anything at all as it is too imprecise.

More generally by clarifying and defining the terms and concepts we are using we can say precisely what we mean.





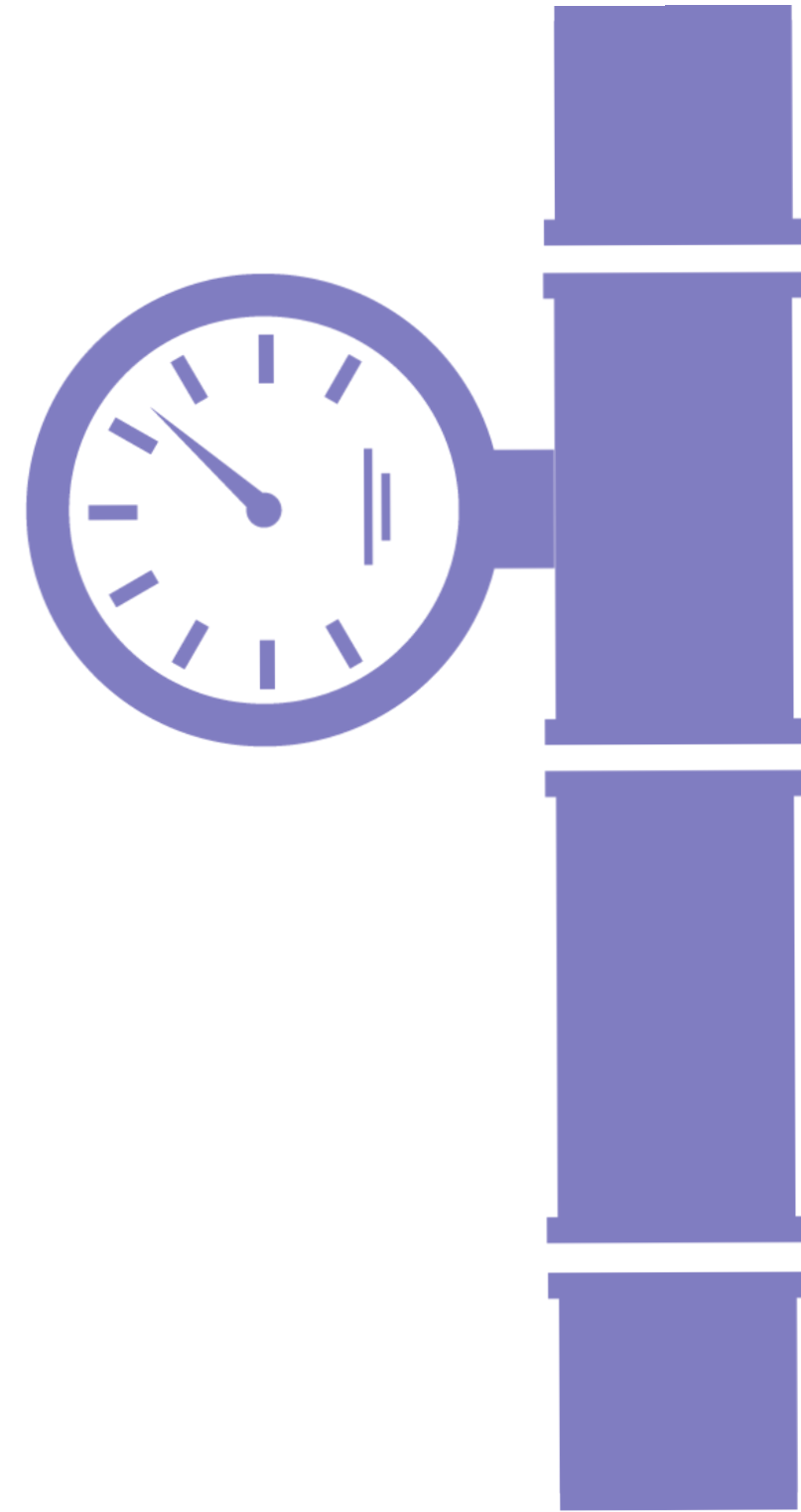
Relevance

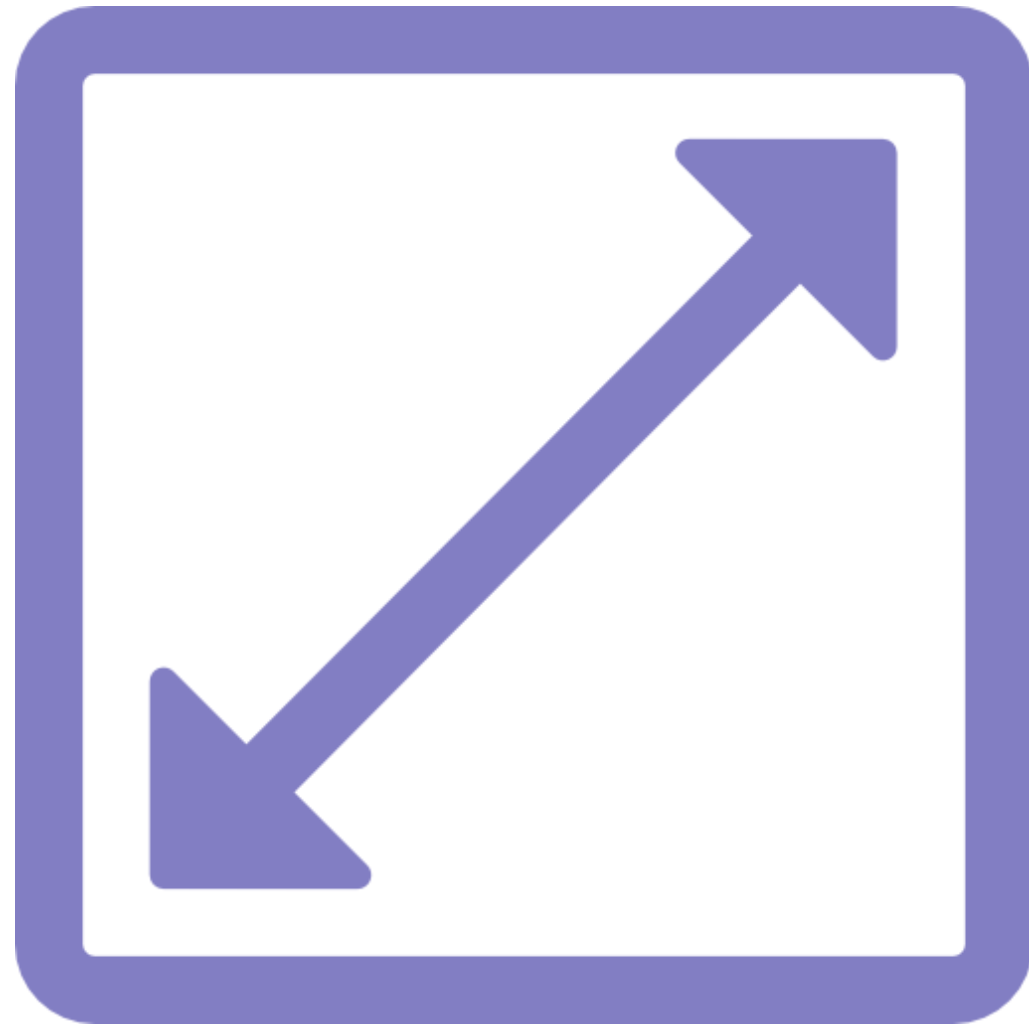
Is what is being said relevant to the topic? Or is the thinking relevant to the problem at hand? Lack of relevance in thought or speak will achieve little and is one way of avoiding the issue. Effective reasoning and dialogue should be adapted specifically to the task at hand and not dealing with some other issue.

Depth

Does one's thinking have sufficient depth to deal with the complexities at hand? Simple questions require only simple answers, complex issues require a depth of insight and thinking.

One should be able to think deeply about an issue if needed, superficial thinking is typically of little value when dealing with a topic of any complexity.





Breadth

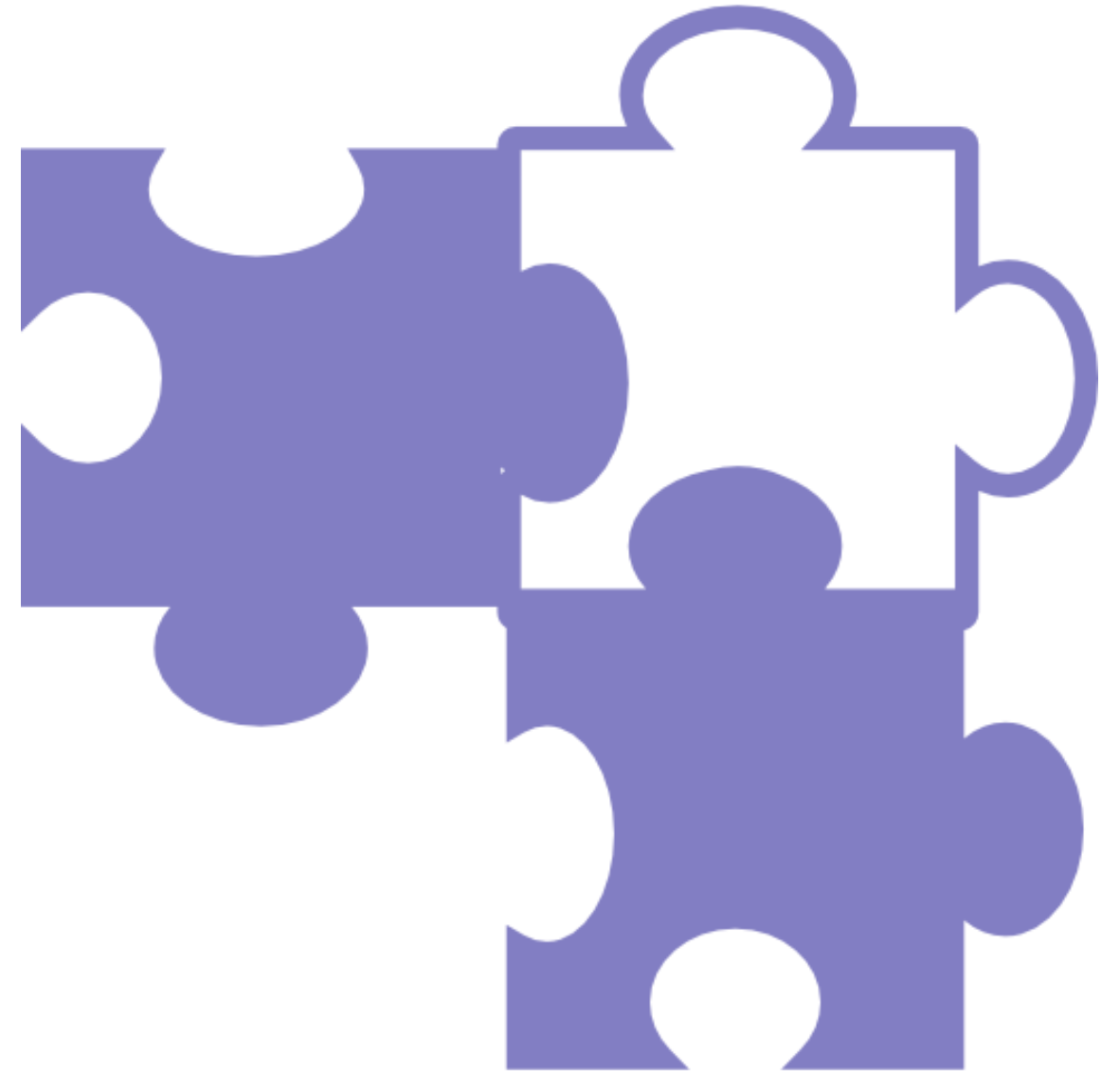
Have we made a full inquiry, gathering data from a wide and diverse set of sources? Is there a sufficient number of perspectives included in the analysis to achieve an inclusive conclusion?

It is possible to go deeply into some specific issue without going horizontally to achieve sufficient breadth of vision and a holistic perspective on the issue.

Logic

Is our reasoning logically consistent? Parts of it may be logically consistent but does it all fit together without inconsistencies?

When the aggregate thoughts are mutually supporting and make sense in combination, then one is thinking in a logical fashion. When the combination in some way is contradictory it is not logical. Our reasoning has to make sense as a whole not just a summation of separate observations or statements.

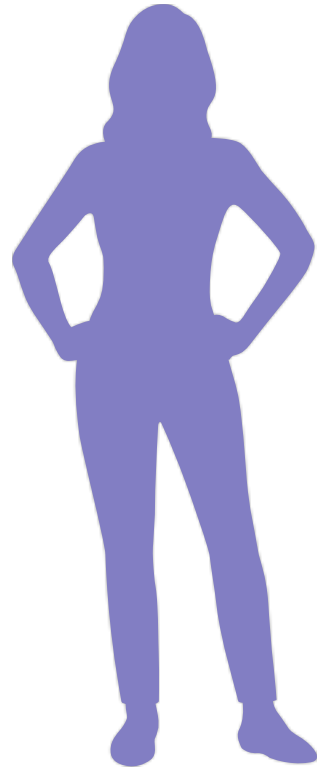




Fairness

Are we conducting an open-minded unbiased inquiry into the situation? Have we determined our desired conclusion before starting the process of reasoning? If so, then it is not truly a fair process.

There are many ways that our own prejudices and self-interests can enter in to manipulate our conclusions toward certain ends, have we given full thought to how that may be present in our reasoning?



Personal Characteristics

Acceptance of Diverse Viewpoints
Desire to be Well Informed
Open and Fair-mindedness
Reflective Skepticism
Inquisitiveness
Seek Reason^[15]



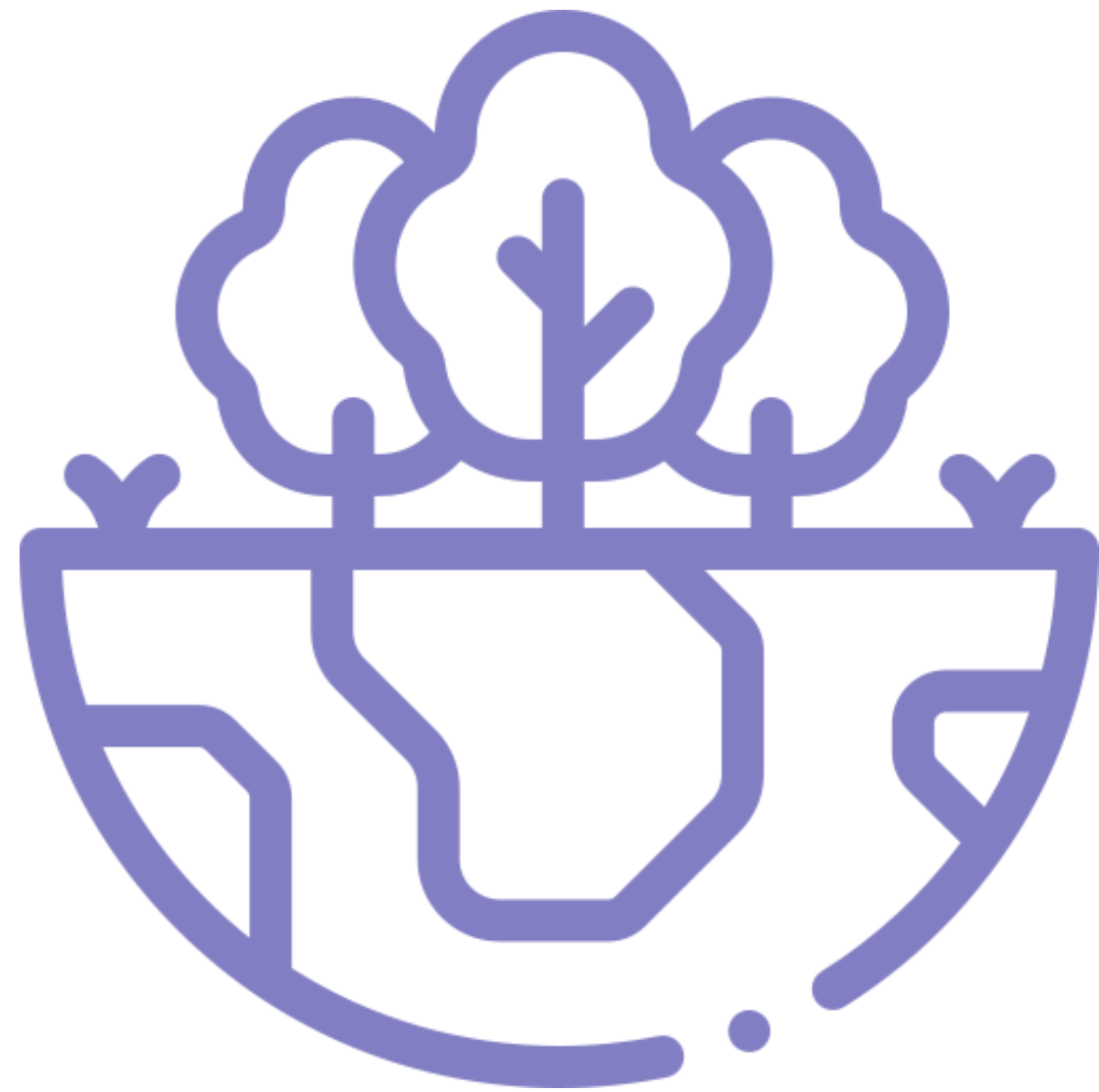
Cognitive Skills

Understand the Links Between Ideas
Determine Relevance of Ideas
Understanding Logic
Able to Build Arguments
Able to Appraise Arguments
Reflect on Justifications
Understand Context^[16]



Capacity to Apply

Willingness to Alter Behavior
Flexible & Adaptive
Act According to Reason
Systematic in Approach
Context Sensitivity^[17]



Ego-system
To
Ecosystem

"I believe the most interesting challenge we face... involves seeing and acting from the whole, and that requires us throughout larger systems to wake up to another level of awareness that we collectively begin to operate from"

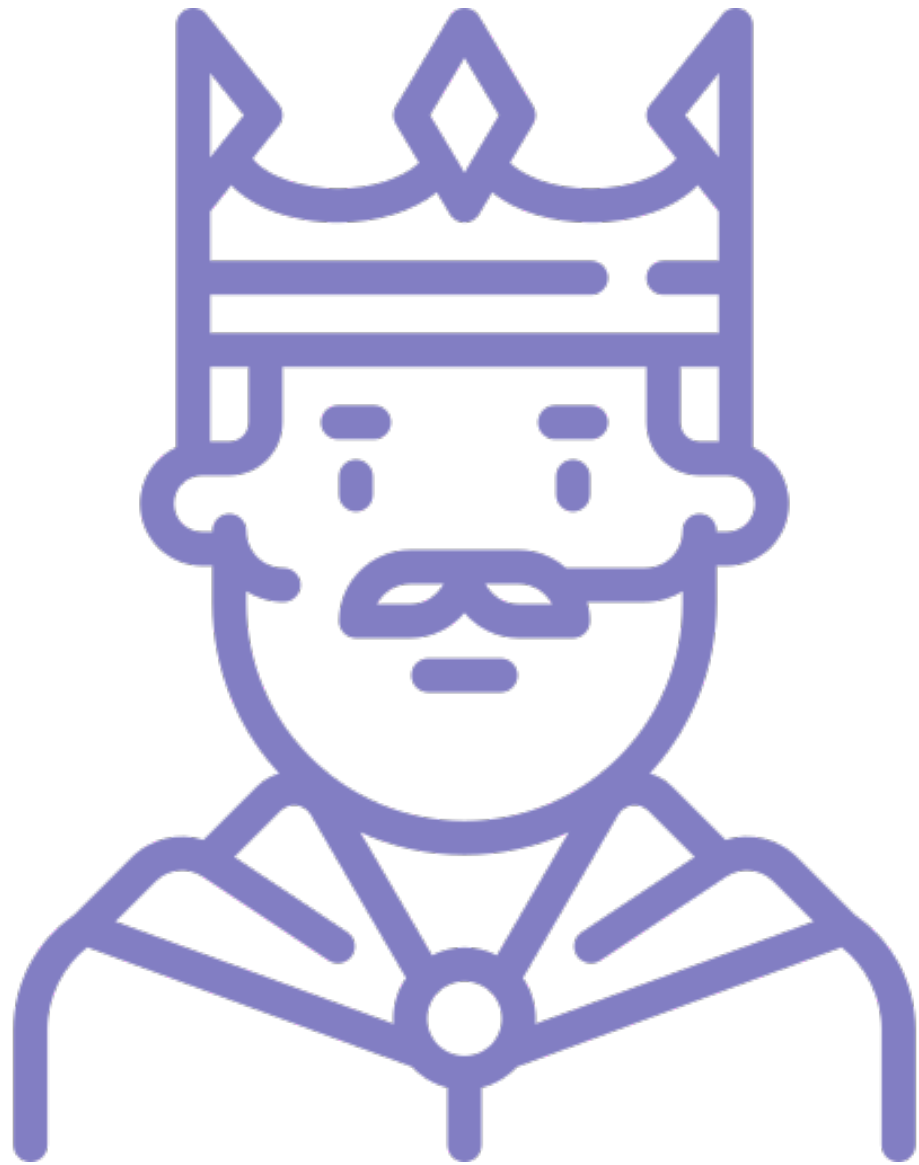
- *Otto Scharmer*

Ecosystems Thinking

It is clear that systems change is going have to come from and be sustained by some new kind of system-level awareness. The inventor of Theory U Otto Scharmer described this shift as a move from ego-system to ecosystem awareness. He sees the current social and economic ways of organizing as fundamentally ego-centered, that is to say that our current organizational and economic structures are designed to satisfy the wants of individuals. We then try to address this issue by extending the awareness of consumers and producers beyond themselves to take in the welfare of other stakeholders. But he sees this as inadequate to deal with the complex sustainability challenges we face, a different kind of awareness is needed; a shift from ego-systems awareness to ecosystem awareness.^[9]



Ego Awareness



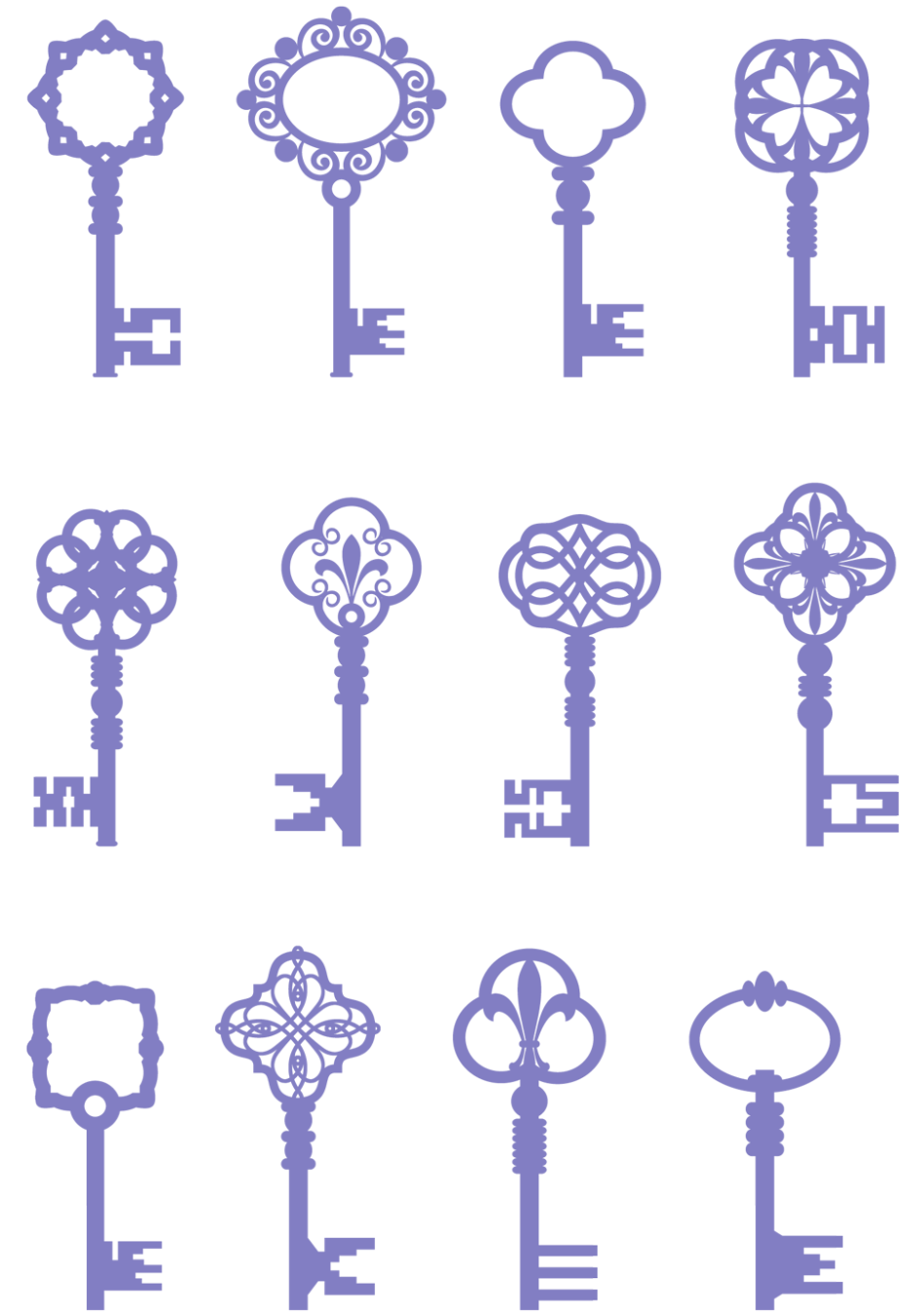
As Otto Scharmer describes it “what’s really needed is a deeper shift in consciousness so that we begin to care and act, not just for ourselves and other stakeholders but in the interests of the entire ecosystem in which economic activities take place.”^[10]

In this respect, it is critical that as systems innovators we become aware of our ego in the change process. As Otto Scharmer notes acting from an ego-centric level of awareness creates a set of motives that will lead us down the wrong path in our initiative. Real sustainable systems change can only be realized and maintained through an awareness of and acting in the interests of the whole systems we wish to change. Unless we address the fundamental structure of our thinking and motives and shift this to an ecosystem-level the same people will create the same problems just in a new structure.

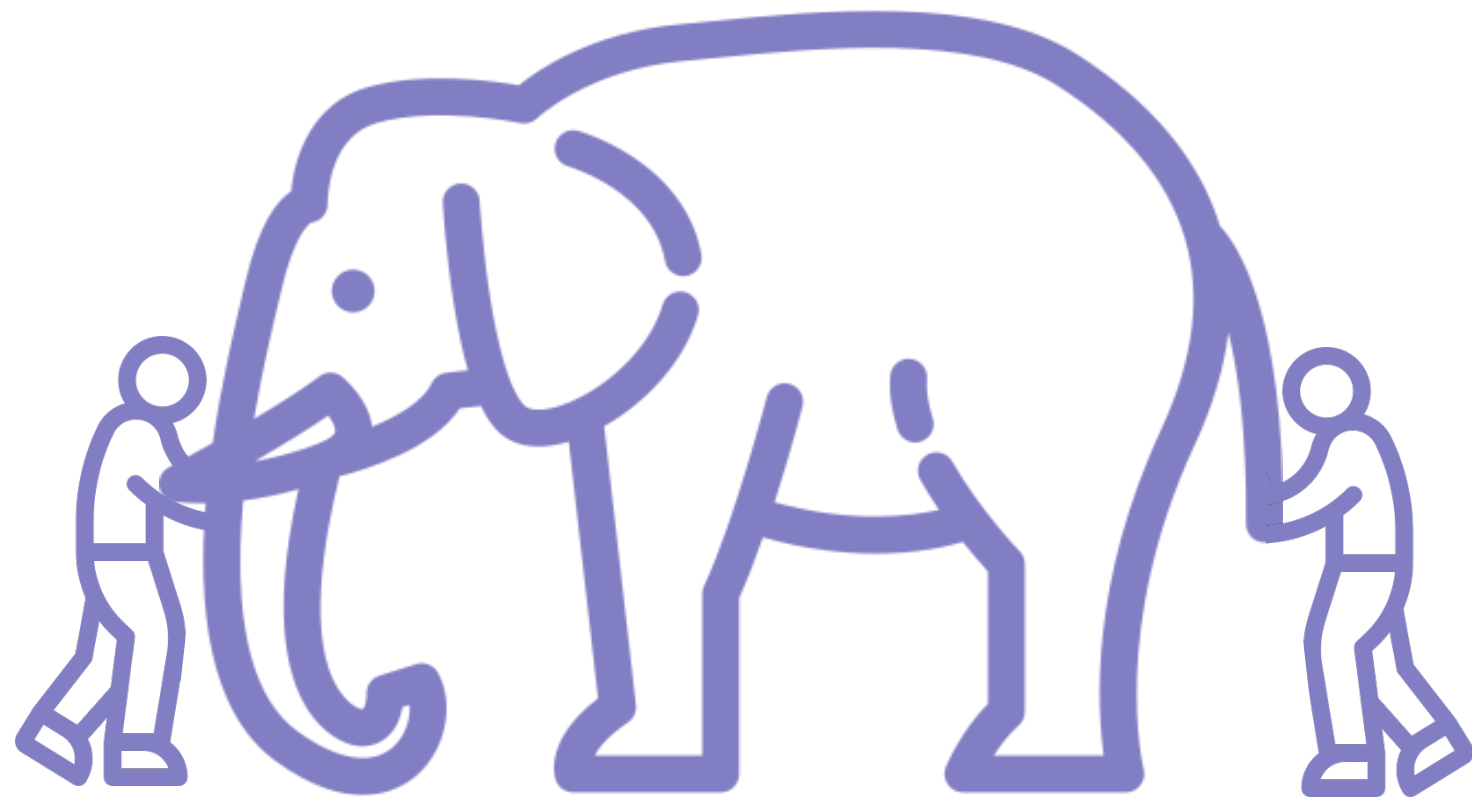
Importance of Perspective

"A systems approach begins when first you see the world through the eyes of another."^[11] This famous quote from Charles Churchman illustrates the need for us to be able to overcome our egocentric view of the world if we wish to become systems thinkers. When we look at the world from a level one kind of thinking we just see it from our own perspective and things in relation to ourselves.

The aim is to be able to actually separate ourselves from our own point of view and start to see the world from different perspectives. When we are able to get beyond our initial level one thinking and see the world from other people's perspective then we can start to piece together a view of the broader systems in our world. Reflexivity and critical thinking are what enable us to get beyond our own perspective to start to value and include multiple perspectives and from them start to see the broader systems we form part of.



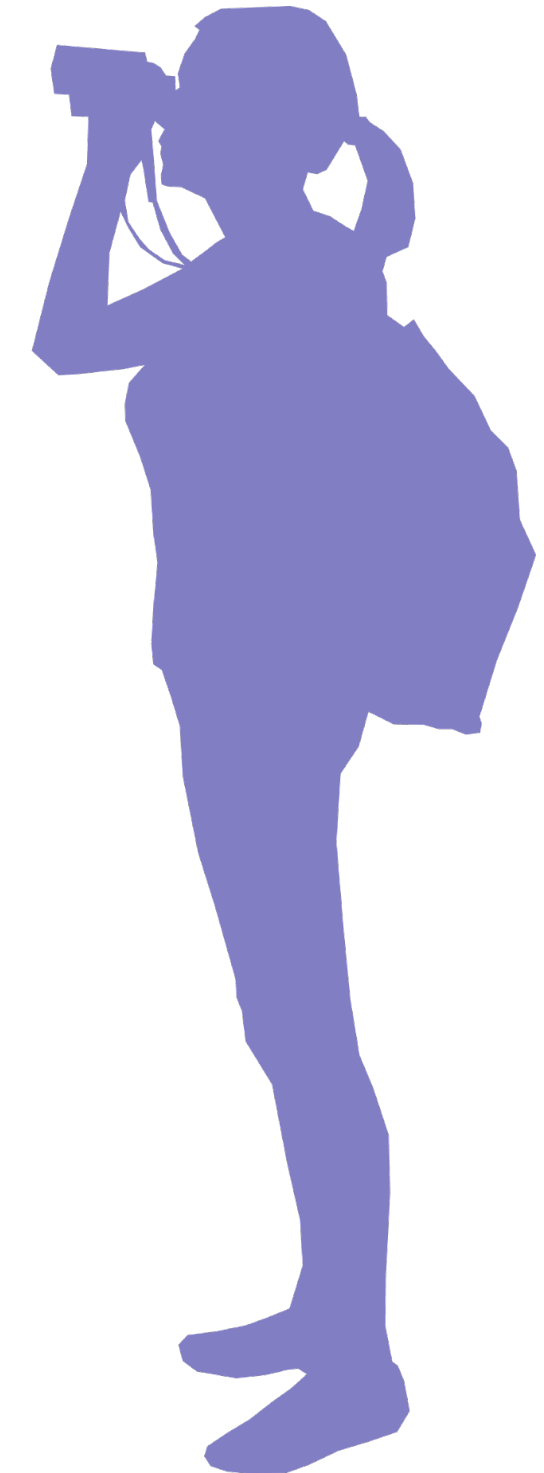
The Elephant & The Blind Men



This recognition of our limited perception and need for systems-level awareness is famously captured in the Indian parable of the blind men and an elephant. This is the story of a group of blind men who have never discovered an elephant before but come to learn about it and try to grasp it conceptually through touching it. Each blind man feels a different portion of the elephant's body, but each only experience just one part, such as the trunk or the hooves. They then describe the elephant based on their limited experience with each creating a different description of the animal. The learning from the story is that people have a tendency to claim complete truth based on their limited, subjective experience while generally ignoring others' perceptions as inaccurate, often resulting in competing positions.^[12]

Seeing the Whole

A primary ability of a systems leader is the capacity to see the larger system and act according to that context. In any complex setting, people typically focus their attention on the parts of the system most visible from their vantage point. This usually results in arguments about who has the right perspective on the problem. It is via the synthesis of different perspectives that we can hope to overcome this and realize systems awareness. Helping people see the larger system is essential to building a shared understanding of complex issues and the job of the systems leader. This is about being better able to listen, see and act within the context of the broader environment the system forms part of.^[13]



Personal Change





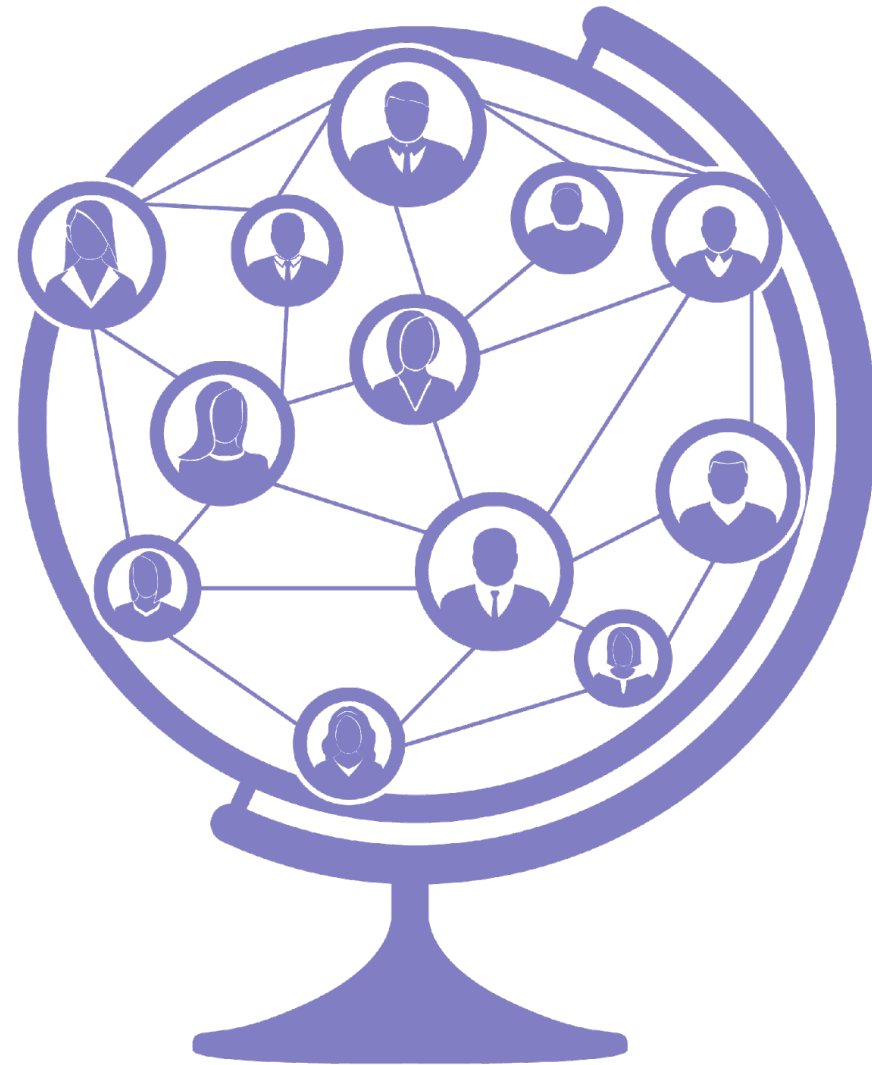
Overview

The first thing to note in any systems change initiative is that change starts with us. As Gregory Bateson said: "you can't change the system without changing yourself." Systems change is certainly about collaboration and the mobilization of whole ecosystems of people and organizations to change systems "out there" however, this process starts with an individual or group of individuals separately feeling something is not right and going on a personal journey of questioning and discovery. That journey inevitably means facing challenges along the way as an individual. Initially, each individual has to face the fears of letting go of the past and exploring the unknown future; systems change can not happen without the subjective personal transformation of those who pioneer it.

Subjective Journey

System change is a deeply personal endeavor, it is as much about the individual's subjective transformation as it is about the objective transformation of the system itself, they are two sides of the same coin, two dimensions of the same process. If we are going to lead change we have to see the new, create it in our lives and live it. Doing this requires that we separate ourselves from what is. Separate ourselves from what we have grown up with and has probably supported us for most of our lives, it requires that we as individuals make a journey of separation and rediscovery, this is an inherently personal and subjective journey.





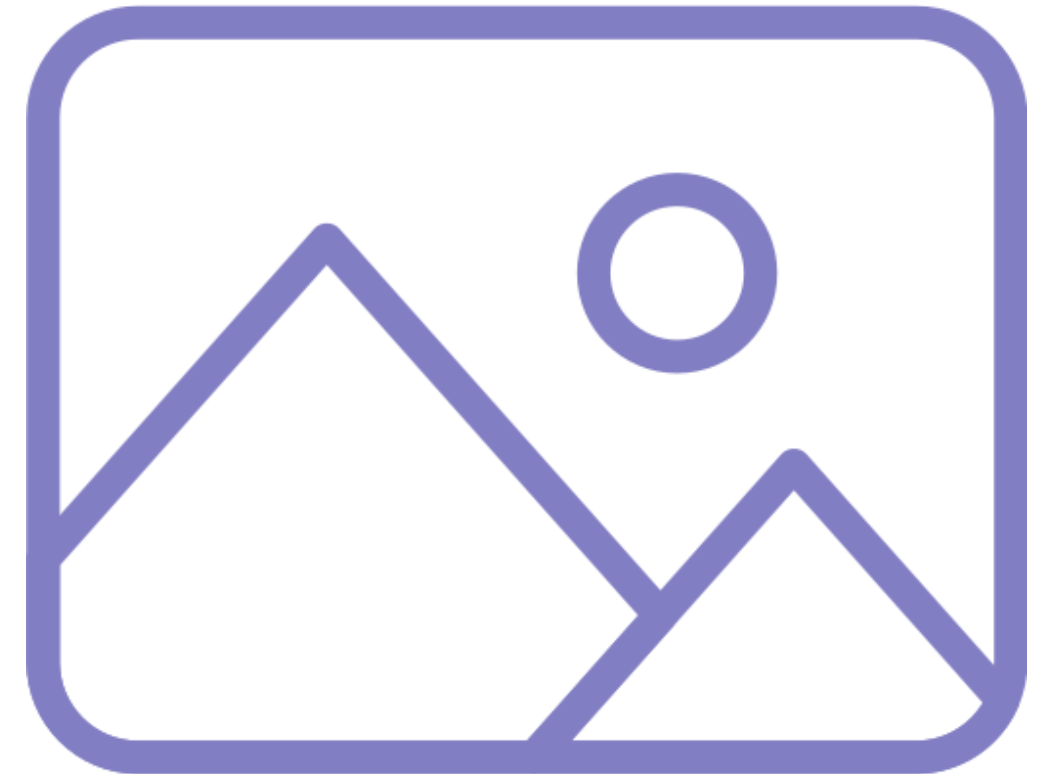
Being Part of the Problem

Likewise to change the system you have to recognize that you are part of it and you are part of the problem. So often our desires to change the system are rendered inert because we separate ourselves from it, posit the problem over there. This is setting the conditions for avoiding systems change, embracing it means embracing our place within the system and our place as part of creating the problem and living with that cognitive dissidence and contradiction within ourselves. We are not the "good guys" the "saviors" we like others are part of the problem and part of working through trying to change the conditions. Real change starts with recognizing that we are part of the systems we seek to change. The failures we seek to remedy also exist within us, as do the fears, distrust, doubt, etc.

Values

Systems change is difficult, but a lot of the difficulty is not out there, it is in one's own subjective journey that begins long before the objective endeavor of changing a system begins. A Lot of the challenge in changing complex organizations is in finding the right motivation and this requires a deep dive exploration into one's values.

Our values are central, we as change agents have to have the right values and act upon them. If you aren't clear about your values and motives your actions will be just as incoherent, and you will dissipate your time and energy going in all directions becoming overloaded. You need to focus on yourself and this focus comes from being clear about what is important and what is not; we come to this conclusion by clarifying our values. This is a process of letting go of those things that are not of value and investing ourselves in those things that are of value.





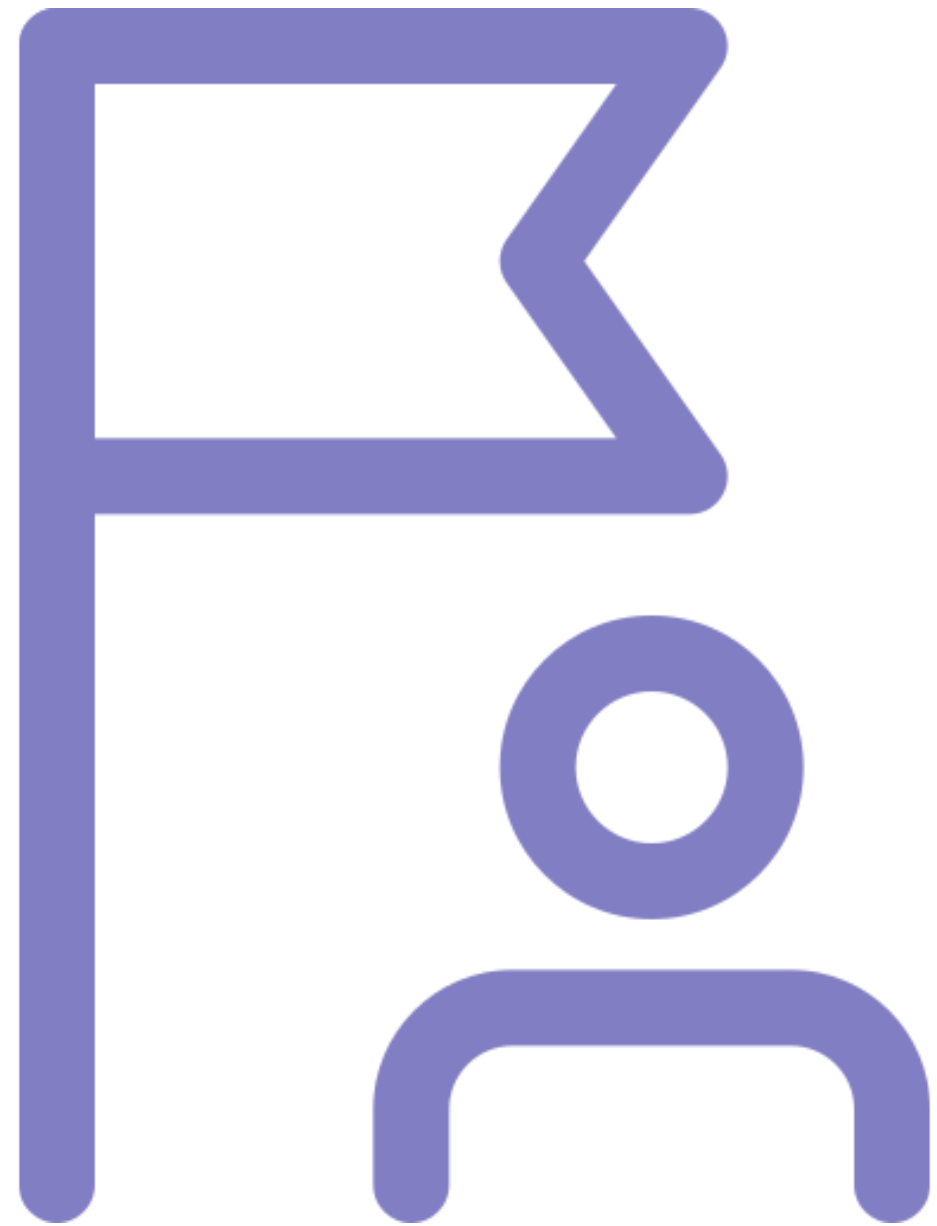
Motivation

Our values are important because they define our motives which ultimately defines how far we can go. Systems change is a creative process and a deeply personal one, people don't change systems due to extrinsic motivations, they do it from deep intrinsic motivation. This intrinsic motivation has to be very strong if they are going to sustain the long term endeavors needed to move a complex organization. Changing a complex system can be a lifetime's work, it takes a strong gritty kind of motivation that comes from something intrinsically at the core of the change agent.

Clarity of Motives

Being clear about your motives and continuously working to expand them is the pathway to success. Being unclear about your motives and having a fixed limit is the road to failure. There really is no such thing as failure when the motives are aligned correctly because you are always prepared to revisit your motives and redefine the scope of your engagement.





Systems Leadership

Overview

Change has to start somewhere. Someone has to start the work of bringing the new possibilities in the system together, at the beginning, this is the work of the leaders. The leaders are the ones who step across the line, create a new space and are prepared to resist the forces that will be inevitably directed at them for challenging and threatening the existing system. Leaders illustrate how, what is, is not the only possible way that things could be, this both inspires people but also threatens the existing regime.

Change often starts with undesirable conditions, but creative system leaders help people move beyond just reacting to these problems to building positive visions for the future. This typically happens gradually as leaders help people express their deeper aspirations, build a shared language, form connections, and build trust based on tangible accomplishments realized together.^[14]



What is Systems Leadership?



To see & act from the whole

People who are aware of the whole context and take actions that are aligned with it



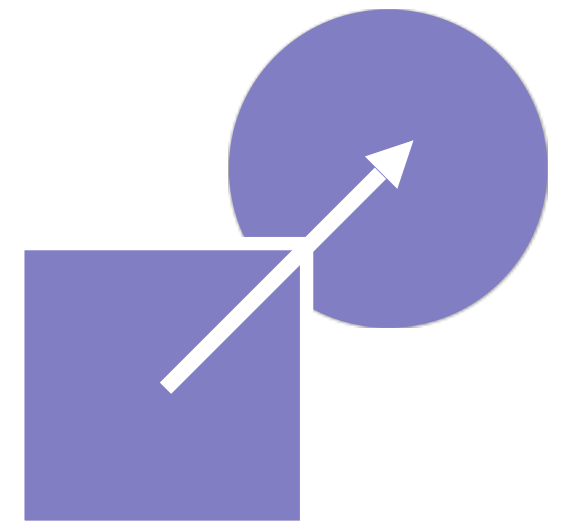
Create conditions for emergence

Leaders empower others, they work with context to create the conditions for emergence



Integrate diversity

People who are working to unite across boundaries rather than divide



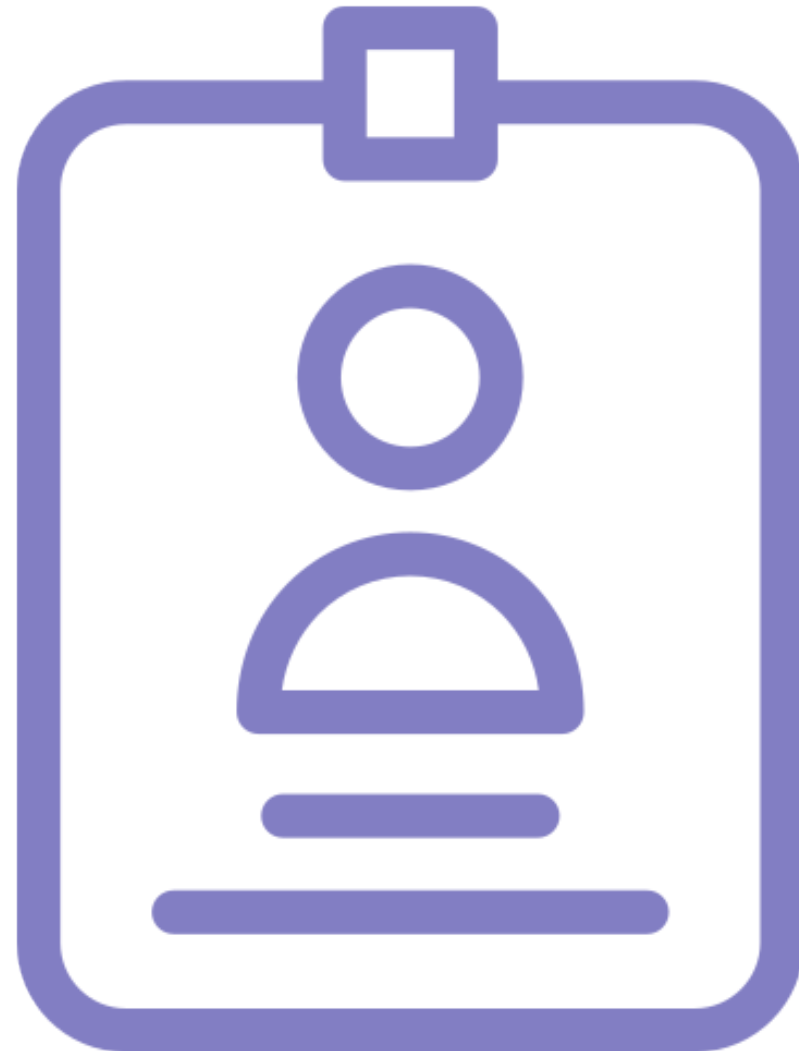
Change Systems

People who see the need to change whole systems rather than just any of the parts



Tip from Banny Banerjee

“When you are addressing really large-scale challenges, and you really are motivated by the big scale, and you want to make a big difference, then what you need are methodologies that are significantly different from current ones and even innovation methodologies that outperform the current innovation methodologies, but also different mindsets, you need different organizational behavior and above all you need a new kind of leadership and that type of leadership we are calling systems leadership and it is significantly different from business as usual”^[15]



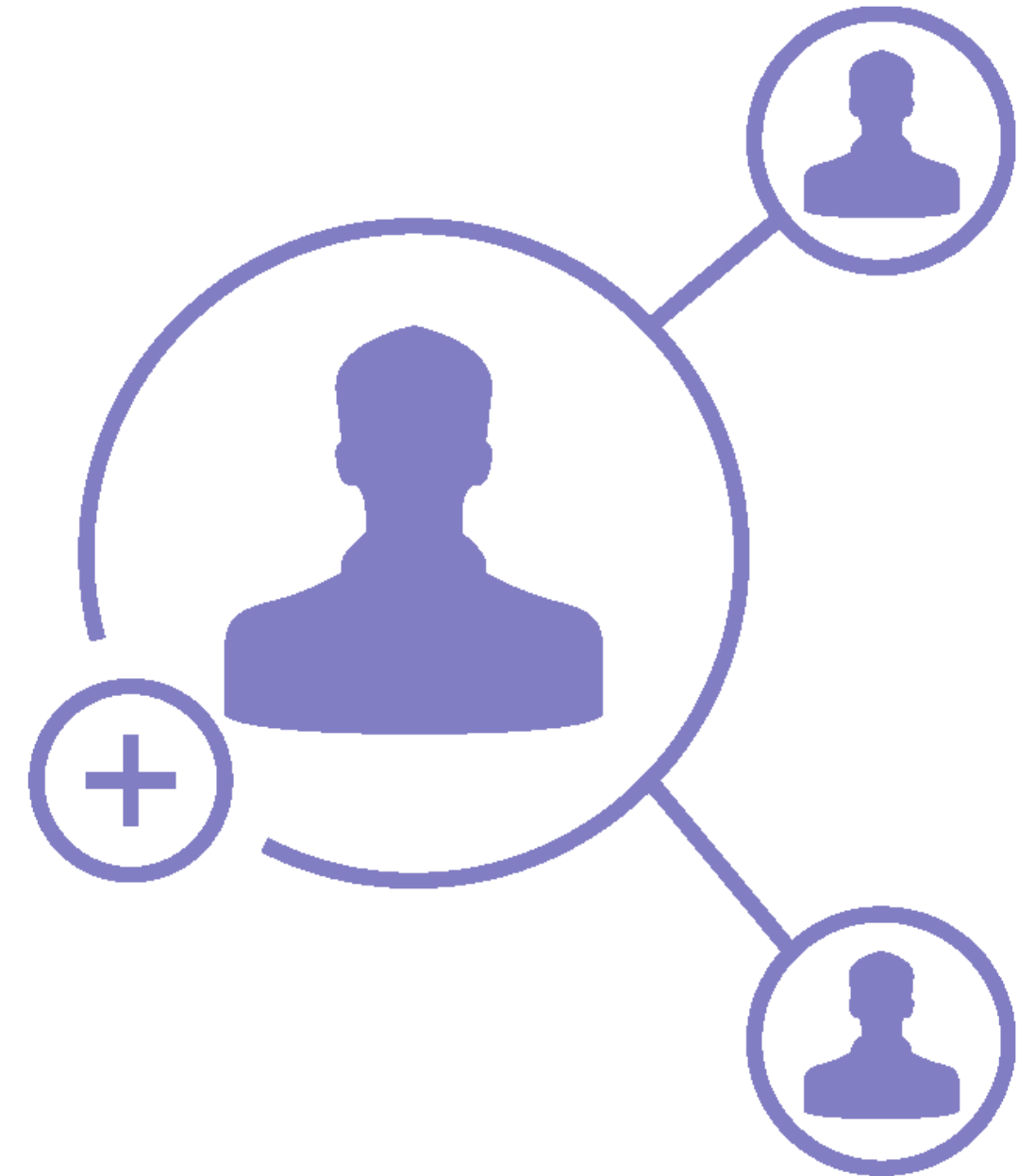
New Kind of Leadership

Because of its nature, the kind of leadership needed for systems change is quite different from our traditional conception. We tend to think of leadership as a position at the top of an organization that comes with the authority to make decisions and have those executed upon by others. However, systems change is about leading across boundaries, meaning you will be leading with very little authority and possibly a role that provides only minimal influence in and of itself. This is leadership as a verb not a noun. It is what we do, not the position we hold that gives us influence.

Leadership as an Action

As Margaret Wheatley notes: "Leadership is anyone who is willing to help, it is not a role its an action... when you see something that needs to be fixed, you step forward, and in that moment you become a leader." Anyone can be a leader by showing leadership qualities.

This is particularly true with systems change because the leaders of the new paradigm are unlikely to come from within the traditional positions of authority, as these are the ones who are most invested in the past and thus least likely to be those to see and create the future. In general, leaders are people who see the whole, the changes taking place within that environment and have a sense that they can shape their future and that others can also - they work to enable groups of people to shape their future.





Responsibility

To be a system changer is to say a better system can be created and you are going to take on some of the responsibility for realizing the change. It requires an appreciation for the multiple dimensions of the system, an appreciation for the complexity that is required to make the system operate on its different levels. As always it is better to not change a system if you don't understand it or have some appreciation for its operations. To lead the change you need to be aware of the system on different levels - technical and nontechnical - and to take on some responsibility for making it work on those different levels.

Reference and Resources

1. Wikiwand. (2020). Awareness | Wikiwand. [online] Available at: <https://www.wikiwand.com/en/Awareness> [Accessed 12 Oct. 2020].
2. Wikiwand. (2020). Pogo (comic strip) | Wikiwand. [online] Available at: [https://www.wikiwand.com/en/Pogo_\(comic_strip\)](https://www.wikiwand.com/en/Pogo_(comic_strip)) [Accessed 12 Oct. 2020].
3. The Schumacher Institute (2014). Introduction to Systems Thinking, Part 1 - How do we view the world? (with Martin Sandbrook). YouTube. Available at: https://www.youtube.com/watch?v=SH94PMHPZW8&t=3s&ab_channel=TheSchumacherInstitute [Accessed 12 Oct. 2020].
4. cxpartners. (2015). Improve your research technique - Reflexive thinking, 5 practical tips | cxpartners. [online] Available at: <https://bit.ly/3jRLAjh> [Accessed 12 Oct. 2020].
5. Criticalthinking.org. (2017). Critical Thinking. [online] Available at: <http://www.criticalthinking.org/> [Accessed 21 Sep. 2020].
6. Criticalthinking.org. (2017). The Elements of Reasoning and the Intellectual Sta. [online] Available at: <http://www.criticalthinking.org/pages/the-elements-of-reasoning-and-the-intellectual-standards/480> [Accessed 21 Sep. 2020].
7. Criticalthinking.org. (2017). The Elements of Reasoning and the Intellectual Sta. [online] Available at: <http://www.criticalthinking.org/pages/the-elements-of-reasoning-and-the-intellectual-standards/480> [Accessed 21 Sep. 2020].

8. Criticalthinking.org. (2017). Universal Intellectual Standards. [online] Available at: <http://www.criticalthinking.org/pages/universal-intellectual-standards/527> [Accessed 13 Oct. 2020].

9. openDemocracy. (2013). From ego-system to eco-system economies. [online] Available at: <https://www.opendemocracy.net/en/transformation/from-ego-system-to-eco-system-economies/> [Accessed 13 Oct. 2020].

10. Odyssey (2019). Otto Scharmer - From Ego-systems to Eco-systems | Odyssey Polaris Keynote. YouTube. Available at: https://www.youtube.com/watch?v=C71mka0Glas&t=329s&ab_channel=Odyssey [Accessed 13 Oct. 2020].

11. Lib Quotes. (2020). C. West Churchman Quote. [online] Available at: <https://libquotes.com/c-west-churchman/quote/lbo2m4h> [Accessed 13 Oct. 2020].

12. Wikiwand. (2020). Blind men and an elephant | Wikiwand. [online] Available at: https://www.wikiwand.com/en/Blind_men_and_an_elephant [Accessed 13 Oct. 2020].

13. The Aspen Institute (2017). Systems Leadership: Tackling Complexity and Scale. YouTube. Available at: https://www.youtube.com/watch?v=HMBT3wkY6KE&t=686s&ab_channel=TheAspenInstitute [Accessed 13 Oct. 2020].

14. Scott.london. (2020). The New Science of Leadership: An Interview with Margaret Wheatley. [online] Available at: <http://scott.london/interviews/wheatley.html> [Accessed 13 Oct. 2020].

15. The Aspen Institute (2017). Systems Leadership: Tackling Complexity and Scale. YouTube. Available at: https://www.youtube.com/watch?v=HMBT3wkY6KE&t=686s&ab_channel=TheAspenInstitute [Accessed 13 Oct. 2020].



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info@systemsinnovation.network